Jaliya Lakshan Wijayaraja

Embedded Systems and AI Researcher | Electronic Engineer | Educator

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ACADEMIC PROFILE

Researcher and educator with a strong engineering foundation in embedded systems, signal processing, and artificial intelligence. Experienced in interdisciplinary research that bridges rigorous scientific investigation with real-world implementation, with contributions spanning electronic engineering, environmental conservation, disaster management, defence, biomedical engineering, and agricultural technology. Author of multiple peer-reviewed publications, including first-author journal articles in Q1-ranked open-access journals and papers in recognised international IEEE conference venues. Recognised through national research and innovation awards, with a research-driven commitment to analytical rigour, technological innovation, and societally relevant impact.

ACADEMIC QUALIFICATIONS

- **Master of Philosophy in Computer Networks**

  Sri Lanka Institute of Information Technology (2025)

  Thesis title: Elephant Detection and Early Warning Using Seismic Waves.

- **Bachelor of Science of Engineering Honours in Electrical & Electronic Engineering**

  Sri Lanka Institute of Information Technology (2020)

  GPA: 3.62 | Second Class Honours in Upper Division.

- **Professional Graduate Diploma in IT (BCS Level 6)**

  BCS Higher Education Qualification (2017)

CERTIFICATIONS

Machine Learning - *Stanford Online, Coursera (May 2021)* [Credential ID: 5QZ9R9QNPP43]

AWARDS, SCHOLARSHIPS, AND GRANTS

Awards and Scholarships

- **Sri Lanka Institute of Information Technology (SLIIT) Research Excellence Award, 2023 and 2024**
  Awarded by SLIIT in recognition of outstanding research contribution among academic staff, reflecting scholarly productivity and academic impact.

- **Outstanding Paper Award, International Conference on Advancements in Computing, 2024**
  Awarded in recognition of the quality, originality, and research contribution of the paper "Understanding Catchment Impacts: Enhanced Water Level Prediction with LSTM and Explainable AI" presented at ICAC 2024.

- **ICT Student Award – Postgraduate Category, Computer Society of Sri Lanka (CSSL National ICT Awards), 2023**
  National-level recognition for outstanding postgraduate research and innovation in information and communication technology.

- **Bronze Award (Postgraduate Category), National ICT Awards – NBQSA Sri Lanka, 2023**
  National-level recognition for postgraduate research and innovation, awarded for JumboSense, an ICT-based solution developed to mitigate human-elephant conflict.

- **IES Student and Young Professionals Paper Assistance for the 49th Annual Conference of the IEEE Industrial Electronics Society.**
  Competitive international support awarded for participation and presentation of the paper "Towards Accurate RPM Detection Using Primary Ignition Pulses" at IEEE IECON 2023.

- **Dean's list recognition for academic excellence – BSc Engineering**
  [1st Yr 1st Sem (2017) | 2nd Yr 2nd Sem (2018) | 3rd Yr 2nd Sem (2019) | 4th Yr 2nd Sem (2020)]
  Awarded in recognition of high academic performance across multiple semesters of the undergraduate engineering program.

Grants

- **SLIIT Research Grant Programme, 2023**
  MPhil research supported under the annual research grant programme of the Sri Lanka Institute of Information Technology, under the Principal Investigator Dr. Janaka Wijekoon.

- **GoviLab AgriTech Accelerator, 2024**
  Farm's Tech (Pvt) Ltd, a startup co-founded to translate research into practical agricultural technologies, received support through the GoviLab AgriTech Accelerator in 2024. The programme, implemented in partnership with Hatch and Sarvodaya under Sri Lanka's Inclusive Digital Agriculture Transformation (IDAT) strategy, supports agri-tech ventures developing scalable, technology-driven solutions for the agricultural sector.

PROFESSIONAL EXPERIENCE

Lecturer **Feb 2026 - Present**
Department of Industrial Management, University of Kelaniya

- Deliver lectures and conduct assessments for the undergraduate module Mobile Computing.
- Contribute to teaching and research activities in embedded systems, IoT, electronics, and machine learning.

Contract R&D Engineer **Sep 2025 - Present**
Independent Subcontractor for CorSci LLC, USA

- Design electronic systems and firmware for biomedical device subsystems, with a focus on reliability and performance optimization.

Co-Founder/ R&D Engineer (Consulting) **Feb 2023 - Present**
Farm's Tech (Pvt) Ltd

Co-founded a technology venture inspired by MPhil research at SLIIT, focused on engineering solutions for the agricultural sector in Sri Lanka. The venture received support through **the GoviLab AgriTech Accelerator** under Sri Lanka's Inclusive Digital Agriculture Transformation initiative.

- Designed and deployed animal repellent systems currently operating across Sri Lanka.
- Leading the development of GoviNet, a comprehensive mobile application for agricultural support and service delivery.

Academic Instructor **May 2025 - Feb 2026**
Faculty of Computing, Sri Lanka Institute of Information Technology

- Conduct undergraduate lab sessions in Control Systems Engineering, Embedded Systems, Robotics and Intelligent Systems, IoT, and Hardware Security.
- Supervised final-year research projects as an external mentor in AI and embedded systems

Research Assistant **May 2021 - May 2025**
Faculty of Computing, Sri Lanka Institute of Information Technology

- Worked as a main researcher on interdisciplinary research into human-elephant conflict mitigation, developing AI-based early warning systems using seismic signal processing, embedded engineering, and machine learning.
- Supported undergraduate and postgraduate teaching activities in Embedded Systems, Robotics and Intelligent Systems, IoT, and Hardware Security.

Resource Person (Visiting Lecturer) **Aug 2022 - Sep 2023**
Metropolitan Campus KDU

- Delivered and developed the specialized course "Certificate in IoT & Big Data" for officers of the Office of the Chief of National Intelligence and Disaster Management Division, Ministry of Defence, Sri Lanka.
- Designed course material focused on IoT and electronics for defense and emergency response applications.

Electronics Engineer (Trainee) **Nov 2019 - Jan 2020**
Centre for Defence Research and Development,
State Ministry of National Security, Home Affairs and Disaster Management.

- Contributed to the development of a digital voice encryption module, defence-grade jammer system design and manufacturing, and a computer-vision application for a T-55 main battle tank training simulator.

June 2020

- Voluntarily contributed to the design of jammer systems for national interest, focusing on PCB and electronic design.

Engineer Trainee **Nov 2018 - Jan 2019**
Vega Innovations

- Contributed to the development of the AIGrow Smart Sense Android application and AIGrow Sensor Node for real-time agricultural monitoring through digital circuit design and firmware development.

RESEARCH EXPERIENCE

Elephant Detection and Early Warning Using Seismic Waves
[MPhil Research | Sri Lanka Institute of Information Technology]

Conducted Master of Philosophy research on a non-invasive embedded sensing and machine learning framework for long-range elephant detection using seismic signals for human-elephant conflict mitigation in Sri Lanka. The study introduced a novel geophone-sensor interface, a Contextually Customized Window (CCW) event detection method, and a classification framework for elephant-generated seismic events. The research achieved detection up to 155.6 m, with classification accuracies of 99% in controlled settings, 73% in natural environments, and 70% in human-elephant conflict-prone village settings. This work resulted in two IEEE Access journal papers and received national-level recognition through the CSSL ICT Student Award and NBQSA Bronze Award.
Role: MPhil Researcher

Evaluation of a Retrieval-Augmented Generation (RAG)-Based System for Automated Viva Assessments in STEM Education
[Ongoing Research]

Leading the development and evaluation of a Retrieval-Augmented Generation (RAG)-based autonomous viva assessment framework for STEM education. The system integrates vector retrieval, prompt engineering, few-shot calibration, and LLM-based reasoning to generate personalised viva questions, evaluate student responses, and support transparent and scalable assessment. The framework was evaluated across multiple academic modules with more than 250 students and has led to an IEEE Access manuscript under review.
Role: Principal Investigator

Machine Learning for Flood Forecasting and Analysis in Sri Lanka
[Government Collaboration]

Conducted research on interpretable machine learning for flood forecasting in Sri Lanka in collaboration with the Disaster Management Division, Ministry of Defence. The study developed an LSTM-based prediction framework integrated with explainable AI and introduced a context-based preprocessing approach for missing rainfall observations. The work resulted in a peer-reviewed conference publication and received the Outstanding Paper Award at ICAC 2024.
Role: Principal Investigator

Improving the Accuracy of Automobile RPM Sensing Using Ignition Pulses from Distributor-Based Ignition Systems.

Investigated signal processing methods to improve RPM sensing accuracy using primary ignition pulses from distributor-based ignition systems. The study proposed a less mechanically intrusive sensing approach and demonstrated that a simple RC filtering method could improve measurement reliability. This work resulted in a publication at IEEE IECON 2023 and was recognized through the IEEE IES Student and Young Professionals Paper Assistance award.
Role: Principal Investigator

AI-Based Diagnostic Model for Forensic Wound Interpretation: Differentiating Lacerations from Sharp-Force Injuries
[Ongoing Collaborative Research]

Contributing to an interdisciplinary study on the application of computer vision and artificial intelligence to forensic wound interpretation, with emphasis on image enhancement, classification, and explainable AI for improved interpretability of diagnostic outcomes. This study is being conducted in collaboration with the Faculty of Medicine, University of Colombo.
Role: Co-Investigator

Encrypted Radio for Defence Forces
[Final-Year Research Project - BSc Engineering (Hons) | Sri Lanka Institute of Information Technology]

Conducted undergraduate research on a secure communication system for defence applications in Sri Lanka under the supervision of the Centre for Defence Research and Development, Ministry of Defence, Sri Lanka, with emphasis on embedded system design and secure communication mechanisms.
Role: Undergraduate Researcher

PUBLICATIONS

Journal Publications
*[2 **first-author** publications in a **Q1**-ranked **open-access** journal]*

J. L. Wijayaraja, J. L. Wijekoon, M. Wijesundara and L. J. Mendis Wickramasinghe, "*Toward Long Range Detection of Elephants Using Seismic Signals: A Geophone-Sensor Interface for Embedded Systems*," in **IEEE Access**, vol. 12, pp. 72210-72223, 2024, doi: 10.1109/ACCESS.2024.3401855.

J. L. Wijayaraja, J. L. Wijekoon and M. Wijesundara, "Event Detection and Classification for Long Range Sensing of Elephants Using Seismic Signals," in *IEEE Access*, vol. 13, pp. 158366-158384, 2025, doi: 10.1109/ACCESS.2025.3607383.

Conference Proceedings
*[4 peer-reviewed **international conference** publications, including 2 as **first author**]*

J. L. Wijayaraja, N. Abhayasinghe and S. Wellalage, "*Towards Accurate RPM Detection Using Primary Ignition Pulses*," **IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society**, Singapore, Singapore, 2023, pp. 1-5, doi: 10.1109/IECON51785.2023.10312638.

J. L. Wijayaraja, N. D. U. Gamage, P. V. N. U. Pallegangoda, N. Priyashantha, W. B. M. R. Weerasekara and P. K. W. Abeygunawardhana, "*Understanding Catchment Impacts: Enhanced Water Level Prediction with LSTM and Explainable AI*," *2024* ***6th International Conference on Advancements in Computing (ICAC)***, Colombo, Sri Lanka, 2024, pp. 480-485, doi: 10.1109/ICAC64487.2024.10851183.

D. Balasingham, S. Samarathunga, A. Bandara, G. G. Arachchige, N. Gamage and J. L. Wijayaraja, "*Enhancing Train Transportation in Sri Lanka: A Smart IOT Based Multi-Subsystem Approach Using MQTT*," ***TENCON 2024 - 2024 IEEE Region 10 Conference (TENCON)***, Singapore, Singapore, 2024, pp. 895-898, doi: 10.1109/TENCON61640.2024.10902958.

L. Jayakody *et al.*, "*Forest Fire Prevention and Detection System*," **2025 IEEE 5th International Conference on Power, Electronics and Computer Applications (ICPECA)***,* Shenyang, China, 2025, pp. 748-753, doi: 10.1109/ICPECA63937.2025.10928745.

Working Papers

[1 journal manuscript under first revision]

A Hybrid LLM-Optimized Framework for Autonomous Viva Assessment Using Retrieval Augmented Generation and Prompt Engineering [*Status: Under first revision at IEEE Access]*

KEY SKILLS

- **Embedded Systems Engineering and Firmware Development** - *Microcontroller-based design, real-time interfacing, and embedded C programming*
- **Artificial Intelligence and Machine Learning** - *Applied ML for signal classification, forecasting, agentic AI, retrieval-augmented generation (RAG), and explainable AI applications*
- **Computer Vision and Intelligent Systems** - *Visual recognition, simulator integration, and OpenCV-based automation*
- **Signal Processing and Simulation** - *Temporal and spectral analysis, digital filter design, event detection algorithm design, seismic signal analysis, signal modelling and simulation using MATLAB*
- **Electronic Design and Simulation**- *Analog/digital circuit design and testing with Proteus and Multisim*
- **PCB Design**- *Schematic layout and board routing using Eagle CAD*
- **3D CAD Modelling**- *Product casing and prototyping using Autodesk Fusion 360*
- **FPGA & Digital Logic Design** - *VHDL-based implementation for digital control and logic systems*
- **Programming Proficiency** - *C, Python, Java, JavaScript, PHP*
- **Academic & Technical Writing** - *Scientific and academic writing in LaTeX*

LANGUAGE PROFICIENCY

- **English**: Fluent
- **Sinhala**: Native

EXTRACURRICULAR ACTIVITIES

- Member of Engineering Research and Innovation Club – Faculty of Engineering, SLIIT
- 1st place in District open Wushu competition - 2014
- Purple belt in Juuoshin Ryu jiu-jitsu.
- Purple belt in Shaolin Martial Arts.
- Member of the Prefects' Guild– Govt. Science College, Matale.

REFERENCES

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